

Artist Specificity beyond a Shared Painting Effect in Text-to-Image Generation

Anonymous authors

Abstract

An image generator can respond strongly to a painter’s name without recovering what distinguishes that painter from others. We test this distinction using Monet, Sisley, Pissarro and Cézanne, 649 digital painting reproductions, and 31 interpretable color, spatial and texture features. A prospective experiment crosses six image models with 14 fixed scenes, a generic oil-painting clause, four painter clauses and an artist-free control, with two repeated requests per cell. Centering the named outputs within each scene removes their common response. We then measure reference-aligned artist variation and use independent repeats to correct the noise bias in artist-geometry error. Scene-paired regressions compare models on the same reference target. All six models show positive reference-aligned variation, although 82.5–95.7% of their global named-versus-free squared feature change is shared across painters. GPT Image 2 matches the reference-aligned amplitude ($\beta = .999$) but retains substantial error in other artist-contrast directions ($D = 1.226$). FLUX.2 Max has the lowest estimated error ($D = .801$), with adjusted advantages over both GPT Image 2.5 variants; the other 13 pairwise differences remain unresolved. These findings persist under common-square measurement and coarse reference-content balancing. Naming thus produces measurable artist-specific responses, but their strength does not establish faithful recovery of reference painter differences.

1 Introduction

“In the style of Monet” is an instruction to distinguish one painter from others. Yet a visibly painterly output need not satisfy that instruction. A common change in color, contrast or surface texture can make images look less photographic while leaving the differences between Monet, Sisley, Pissarro and Cézanne poorly reproduced. Even a generator that produces distinguishable outputs for all four names may distinguish them along the wrong visual axes. The central question is therefore whether painter-name prompting recovers *artist-specific variation*, beyond the appearance shared by its named outputs.

Distance to a painter’s collection alone cannot settle this question. A generic painting transformation could lower that distance for several painters at once. Conversely, correct differences between painters could coexist with a common offset from all reference paintings. These are distinct errors with different implications for model comparison. A permutation of painter labels illustrates the problem: generated collections can remain separable while their names map to the wrong reference painters. An evaluation should separate the common response, the amplitude of reference-aligned differences, and artist contrasts that depart from the reference geometry.

We make that separation experimentally and statistically. Six generators receive the same scene descriptions with either no painting clause, a generic oil-painting clause, or one of four painter clauses. We subtract the mean response across painters within each model and scene, then compare the remaining painter contrasts with those in a fixed reference panel. Two repeated requests permit cross-product estimation of conditional mean error without its usual additive sampling-noise bias. An additive feature shift shared by all four painter conditions cancels from this endpoint.

This design also gives a limited but testable answer to *why* generated collections differ. A shared painting shift, weak or exaggerated artist contrasts, differences along other feature axes, and dependence on the depicted scene produce different signatures in the decomposition. Generic-clause and palette interventions probe alternatives to a uniquely artist-specific response. These are explanations of observable output behavior. They do not identify training frequencies, guidance rules or the internal representation of a painter.

The contribution is a controlled test of reference-relative artist specificity, with a common target for comparing six current model configurations. It combines painter contrasts, generic controls and repeat-corrected error in an interpretable feature space. We retain distributional proximity and spread as complementary outcomes: reproducing differences among painter means is a narrower achievement than reproducing the painters’ full distributions.

2 Related work

Artist imitation and computational style measurement are established research problems. [Somepalli et al. \(2024\)](#) develop learned style descriptors and study style similarity across diffusion models. [Su et al. \(2025\)](#) evaluate prompted-artist recognition using content-controlled name substitutions; they also compare named and artist-free images with real-artist prototypes in CLIP space. Their work already establishes that artist names can leave recognizable signatures. Our question is whether those signatures recover the *relative geometry of the reference painters*, after removing the shared response. Neither name substitution nor cross-model comparison is claimed as a new method here.

[Deliège et al. \(2025\)](#) examine stylistic imitation and stereotyping with expert assessments of historical and generated art. AI-Pastiche studies authenticity judgments and prompt adherence ([Asperti et al., 2025](#)); AI-WikiArt uses painting-derived captions and vision-language models to study attribution ([Fu et al., 2025](#)). These studies motivate distinguishing convincing imitation, recognizable authorship and distributional recovery. We test reference-relative painter differences and summarize full-distribution outcomes in explicit digital coordinates, without substituting them for expert judgments.

Quantitative art research has examined color, composition and image statistics across painting collections ([Kim et al., 2014](#); [Sigaki et al., 2018](#); [Lee et al., 2020](#)). [Kim et al. \(2026\)](#) combine formal and contextual representations to study art-historical change, including contextual image-to-image experiments. Our intervention instead adds painter names to text-only scene prompts, with no original composition supplied to the generator. Related content and style remain entangled in our reference panel; the common-scene intervention controls the requested content of generated images, not the content of historical works.

Generative-model evaluation distinguishes fidelity and diversity ([Naeem et al., 2020](#)), and conditional metrics distinguish variation within conditions from relationships among their means ([Benny et al., 2021](#)). Our cross-repeat estimator follows the established principle of estimating squared signal differences from independent partitions ([Diedrichsen and Kriegeskorte, 2017](#)). Energy statistics provide the complementary distributional endpoint ([Székely and Rizzo, 2013](#)). Processing can materially change image comparisons ([Parmar et al., 2022](#)); separation in a learned embedding is likewise not itself perceptual validation ([Asperti, 2026](#)). Together, these precedents motivate a test of reference-relative artist contrasts with generic controls and independent repeats. The contribution is the combined experimental test and its findings, rather than a new style metric.

3 Design and measurements

3.1 Painters and reference images

The primary panel comprises 649 distinct painting reproductions: 297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne. Artist means receive equal weight, so the larger Monet collection does not dominate the between-artist target. A separate development panel of 221 works supplies feature scaling; no primary reference or generated image is used to fit that transformation. Both panels were assembled before the present experiment and had been used in earlier analyses. They are explicit, finite comparison panels rather than newly withheld samples of each painter’s complete oeuvre.

Reference files are linked to recorded work identities and Wikimedia Commons source metadata. Commons supports traceable file histories, source and rights fields, and institutional collection contributions through its cultural-heritage partnerships (Wikimedia Commons, 2026). Wikidata supplies structured, reusable work and artist identities (Vrandečić and Krötzsch, 2014). These properties make the selection and provenance inspectable; they do not guarantee uniform photography, color calibration or conservation state. We therefore call the inputs *digital reproductions*, not direct measurements of physical paintings. Previous computational art corpora support the use of documented reproductions for comparative analysis, but do not authenticate this collection: the Web Gallery of Art in Kim et al. (2014), ART500K in Kim et al. (2026), and WikiArt in Fu et al. (2025) are distinct from Wikimedia Commons.

3.2 A common-scene, six-model experiment

We compare GPT Image 1, GPT Image 2, GPT Image 2.5 Flare, GPT Image 2.5 Sunburst, Nano Banana 2 and FLUX.2 Max. Fourteen authored outdoor scenes span water, built, land and mixed compositions. Each scene is crossed with six clauses: no painting instruction; “Render as an oil painting”; and “Render as an oil painting in the style of [painter]” for each of the four painters. The scene text is otherwise identical. Each model–scene–clause cell receives two separately requested outputs, giving 1,008 images collected on 10 September 2026.

All models receive text only and return 1,024-by-1,024-pixel images. The four OpenAI models use medium quality; the other two use their specified 1K configurations. These comparisons concern the recorded configurations, not maximum attainable quality or equal expenditure per image. The exact prompts, parameters and randomized request order were fixed before collection. The 14 scenes are a fixed design, not a random sample of all possible content.

3.3 What the features measure

Images undergo orientation correction and conversion to sRGB using valid embedded profiles; compatible unprofiled files are treated as sRGB. The primary pipeline preserves aspect ratio at a 512-pixel short side, using Lanczos resizing without upsampling. Each of the 31 coordinates is centered and scaled by its median and interquartile range in the development panel, with equal total weight per painter. Thus a unit is a development-panel IQR in that coordinate, not a perceptual just-noticeable difference. The Euclidean metric weights standardized coordinates equally, without further equalizing feature families.

Table 1 explains the representation. Color features summarize lightness and chroma, concentration of hue, and color changes across spatial scales; CIEDE2000 supplies the color-difference calculation (Sharma et al., 2005). Spatial features describe how luminance contrast is organized across frequencies, directions and regions. Texture features summarize multiscale detail and local intensity patterns using wavelets and local binary patterns (Mallat, 1989; Ojala et al., 2002). For example, similar mean chroma can coexist with different hue diversity, and similar

total contrast can coexist with different edge orientation or fine-scale texture. These distinctions make family-specific results interpretable, while correlated coordinates and processing sensitivity prevent treating the Euclidean metric as a validated measure of artistic style.

4 Separating common appearance from artist recovery

4.1 Reference-aligned variation

Let $\mu_a \in \mathbb{R}^{31}$ be the standardized reference mean for painter a and let $\bar{\mu} = \frac{1}{4} \sum_a \mu_a$. Define the reference painter contrasts $r_a = \mu_a - \bar{\mu}$ and their total squared magnitude $H = \sum_a \|r_a\|^2$. For generated vector z_{msak} from model m , scene s , painter a and repeat $k \in \{1, 2\}$, remove the common named response:

$$d_{msak} = z_{msak} - \frac{1}{4} \sum_b z_{msbk}. \quad (1)$$

Any shift shared by all four named outputs of that model, scene and repeat cancels exactly. This includes a common painting effect and any model-specific baseline that acts additively in these coordinates.

The reference-aligned slope is

$$\hat{\beta}_{msk} = \frac{\sum_a \langle d_{msak}, r_a \rangle}{H}, \quad \hat{\beta}_m = \frac{1}{S} \sum_s \frac{\hat{\beta}_{ms1} + \hat{\beta}_{ms2}}{2}. \quad (2)$$

It is the least-squares coefficient of generated painter contrasts on reference contrasts. A value of zero indicates no response along the reference direction; one matches its amplitude. Values above one indicate exaggeration along that direction, not necessarily better recovery. A nonzero slope also does not exclude substantial differences along other axes.

4.2 Noise-corrected error and its explanation

Two independently requested repeats permit the cross-product estimator

$$\hat{D}_{ms} = \frac{1}{H} \sum_a \langle d_{msa1} - r_a, d_{msa2} - r_a \rangle, \quad \hat{D}_m = \frac{1}{S} \sum_s \hat{D}_{ms}. \quad (3)$$

Family	Measurements	Interpretation
Color (11)	Lightness and chroma median/IQR; chromatic fraction; hue concentration and entropy; color differences at three lags and their slope	Brightness, color strength and diversity, and how rapidly colors change over space.
Spatial (8)	Spectral slope, residual and anisotropy; orientation entropy; horizontal-vertical balance; gradient median/IQR; quadrant divergence	Coarse versus fine organization, dominant directions, edge strength and regional variation in orientation.
Texture (12)	Four wavelet energies, slope and curvature; three local-binary-pattern entropies; three local coefficients of variation	Detail across scales, diversity of local patterns and local luminance contrast.

Table 1: The 31 fixed digital features. Counts refer to scalar coordinates. Texture measurements do not measure physical paint relief or authenticate brushwork. Exact scales and definitions appear in Appendix A.

Under stable conditional means and independent, mean-zero repeat errors, its expectation is the squared error of the scene-conditional mean painter contrasts, normalized by H . The error contributed by output sampling does not inflate this expectation. Exact mean-geometry recovery gives expected $D = 0$; a purely common named response gives expected $D = 1$. Finite estimates may be negative. We do not truncate them or interpret a negative estimate as better-than-perfect recovery.

The estimator has an exact decomposition. Write $e_{msak} = d_{msak} - \hat{\beta}_{msk}r_a$. Then

$$\hat{D}_{ms} = (\hat{\beta}_{ms1} - 1)(\hat{\beta}_{ms2} - 1) + \frac{1}{H} \sum_a \langle e_{msa1}, e_{msa2} \rangle. \quad (4)$$

The first term is error in the amplitude of the reference-aligned response; the second is error orthogonal to that direction. This distinguishes weak or exaggerated conditioning from a different organization of painter contrasts. Both terms retain the cross-repeat correction and can be negative in finite samples. Computing the same error after averaging over scenes supplies a separate descriptive target: low aggregate error can conceal scene-dependent failures.

4.3 Comparing models

We fit the scene fixed-effects regression

$$\hat{D}_{ms} = \alpha_s + \gamma_m + \varepsilon_{ms}, \quad \gamma_{\text{GPT Image 2}} = 0. \quad (5)$$

All models face the same scenes and equally weighted reference painter target. With a complete balanced panel, model coefficient differences equal the mean paired scene differences. Inference uses those scene differences rather than treating features, painters or individual images as independent replicates.

The fixed inferential family contains six reference-aligned slopes and 15 pairwise model error contrasts. We report paired-scene Student intervals with $n - 1$ degrees of freedom and Bonferroni adjustment across these 21 comparisons. Nominal 95% intervals for absolute D_m are descriptive. These are approximate summaries of variation across the fixed briefs under the assumed error model; randomized request order does not establish independence or justify inference to arbitrary scenes. All 14 scenes are complete for all models.

Prespecified sensitivity analyses remove each scene in turn, resample scenes for model contrasts, and resample reference works within painter while holding generated vectors fixed. The last analysis assesses dependence on this reference panel; it is not a second independent uncertainty estimate. We also repeat the analysis in each feature family, without texture, and using central square measurement windows for both generated and reference images. The square view controls measurement-window aspect ratio while discarding peripheral content. These analyses are descriptive and do not select a preferred representation.

A separately declared sensitivity addresses unequal subject mixtures in the references. Before inspecting the new feature outcomes, we specified equal weight for four previously recorded, metadata-derived content classes within each painter, then recomputed the reference contrasts and all model estimates. For example, water scenes are much more frequent in the Monet panel than in the Pissarro panel. Equal class weighting tests dependence on that broad mixture, without matching individual compositions or identifying pure style. Reference resampling within artist/class strata assesses the stability of this alternative target.

4.4 Generic controls and full distributions

The mean change from the free arm to the four named arms separates into a common component and centered painter contrasts. We compare the direction and magnitude of the common

component with the free-to-generic change. Their cosine similarity and cross-repeat squared magnitudes test whether a generic painting clause reproduces the shared direction. A large common fraction alone cannot show that naming fails: these four reference painters also share visual properties. Equations 2–3 supply the separate recovery test.

For painter a , energy discrepancy compares its generated collection Y_a with reference collection X_a :

$$\hat{\mathcal{E}}(X, Y) = \frac{2}{n_X n_Y} \sum_{i,j} \|x_i - y_j\| - \frac{1}{n_X^2} \sum_{i,i'} \|x_i - x_{i'}\| - \frac{1}{n_Y^2} \sum_{j,j'} \|y_j - y_{j'}\|. \quad (6)$$

Unlike distance to a mean, this statistic incorporates within-collection variation. Its empirical baseline need not be zero for two independent samples from the same distribution. We also report generated/reference total-variance ratios. These complementary outcomes can reveal narrow or displaced generated distributions even when reference painter contrasts are recovered. We also report within-scene repeat spread, $\frac{1}{2S} \sum_s \|z_{msa1} - z_{msa2}\|^2$, the mean sample-variance trace of the two repeated outputs. This separates repeat dispersion from total variation across briefs.

5 Results

5.1 Aligned artist responses coexist with substantial recovery error

All 1,008 outputs yield valid measurements, completing the 14-scene panel for each model. Every reference-aligned slope has a simultaneous interval above zero (Table 2). Painter names therefore produce a response beyond a purely shared shift. Yet response amplitude does not track recovery error. GPT Image 2 nearly matches the reference amplitude ($\beta = .999$), while its corrected error is $D = 1.226$. The decomposition attributes essentially all of that estimate to other artist-contrast directions (1.229), with a near-zero amplitude-error term ($-.003$). A strong, correctly sized component can coexist with substantial misalignment.

FLUX.2 Max has the smallest estimated error, .801, despite a weaker aligned response ($\beta = .470$). The paired-scene regression resolves two differences after the fixed 21-comparison adjustment: Flare minus FLUX is .937 [.256, 1.618], and Sunburst minus FLUX is .840 [.058, 1.623]. The remaining 13 model differences include zero, including FLUX versus GPT Image 2 and Nano Banana 2. Thus the evidence supports an advantage over the two GPT Image 2.5 configurations on this endpoint, not a uniquely best generator. Even FLUX’s nominal absolute-error interval, [.586, 1.016], includes the purely common-response benchmark of one.

Model	Reference slope β	Error D	ΔD versus GPT Image 2
GPT Image 1	0.938 [0.709, 1.168]	1.572	0.346 [−0.400, 1.093]
GPT Image 2	0.999 [0.893, 1.104]	1.226	0 (baseline)
GPT Image 2.5 Flare	0.772 [0.680, 0.864]	1.738	0.512 [−0.126, 1.150]
GPT Image 2.5 Sunburst	0.824 [0.680, 0.968]	1.641	0.415 [−0.350, 1.181]
Nano Banana 2	0.442 [0.256, 0.629]	1.109	−0.116 [−1.155, 0.922]
FLUX.2 Max	0.470 [0.239, 0.701]	0.801	−0.425 [−1.011, 0.161]

Table 2: Reference-relative recovery and scene-paired model comparisons. Lower D indicates less conditional artist-geometry error; larger β is not necessarily better. Brackets give approximate simultaneous intervals from the fixed family of six slopes and 15 model contrasts. On a complete common panel, model differences equal the scene fixed-effects regression contrasts. Inference uses paired scene differences.

Figure 1 shows the distinction between producing artist contrasts and reproducing their reference arrangement. Its axes are fitted to reference contrasts only; all numerical comparisons use 31 coordinates.

5.2 Shared appearance and off-axis differences explain different effects

The common component accounts for 82.5–95.7% of the cross-repeat corrected global named-versus-free squared feature change. Generic/common direction cosines range from .693 to .888, showing a substantial shared direction. The generic shifts are nevertheless smaller than the common named shifts in every model, with nonzero residuals: a generic clause reproduces part of the shared response, not all of it. A large shared fraction is compatible with positive artist-specific slopes because these statistics answer different questions.

For all four OpenAI configurations, off-axis error dominates the decomposition (Figure 2a). This is not simply greater variation between repeated outputs: the repeat correction removes additive sampling-error bias under its assumptions, and both GPT Image 2.5 configurations have lower within-scene repeat spread than GPT Image 2 for every painter. Scene averaging also conceals some error. Nano Banana 2 has lower global error than FLUX (.723 versus .761), but higher scene-conditional error (1.109 versus .801). Their point ordering reverses when scenes are pooled; the primary paired difference remains unresolved.

5.3 Full distributions and sensitivity qualify the model comparison

All 24 model–painter collections have lower measured total variance than their references, with trace ratios .248–.877 (Figure 2b). Naming lowers empirical reference energy relative to the generic clause in 21 of 24 cells. The exceptions are Flare for Monet and Pissarro, and Sunburst

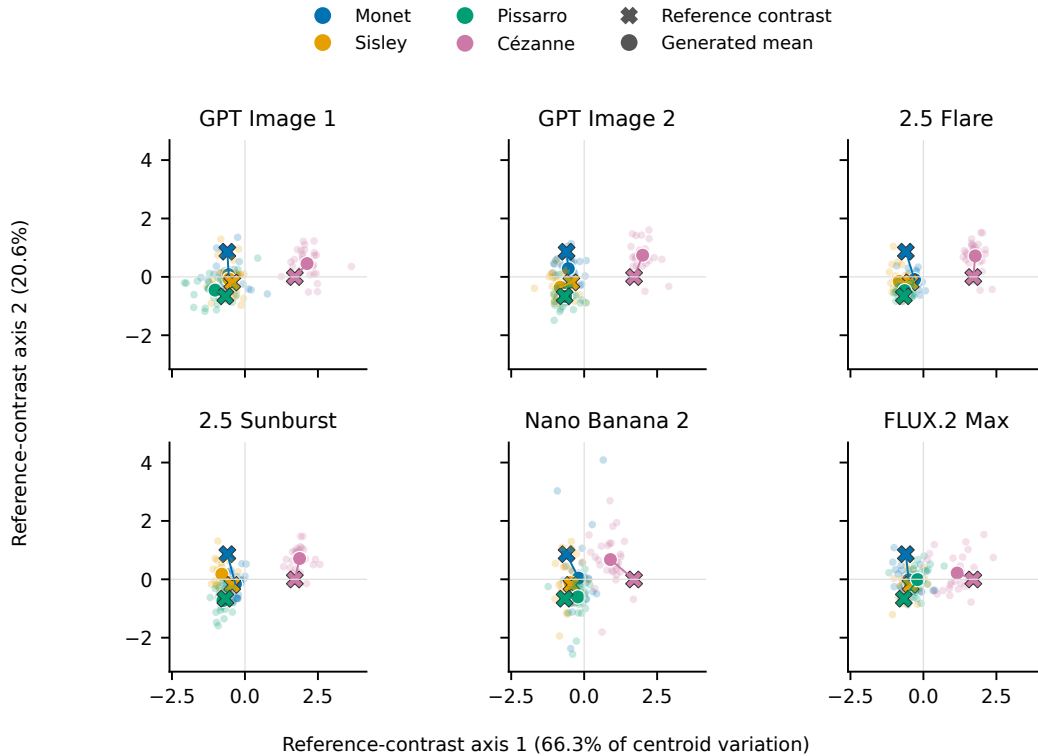


Figure 1: Painter contrasts after removing each scene’s common named response. Colors identify painters; faint points are generated contrasts, circles their means, and crosses the reference contrasts. All panels share axes fitted to the four reference centroids. The projected separation of painter conditions does not by itself establish their alignment with the reference arrangement.

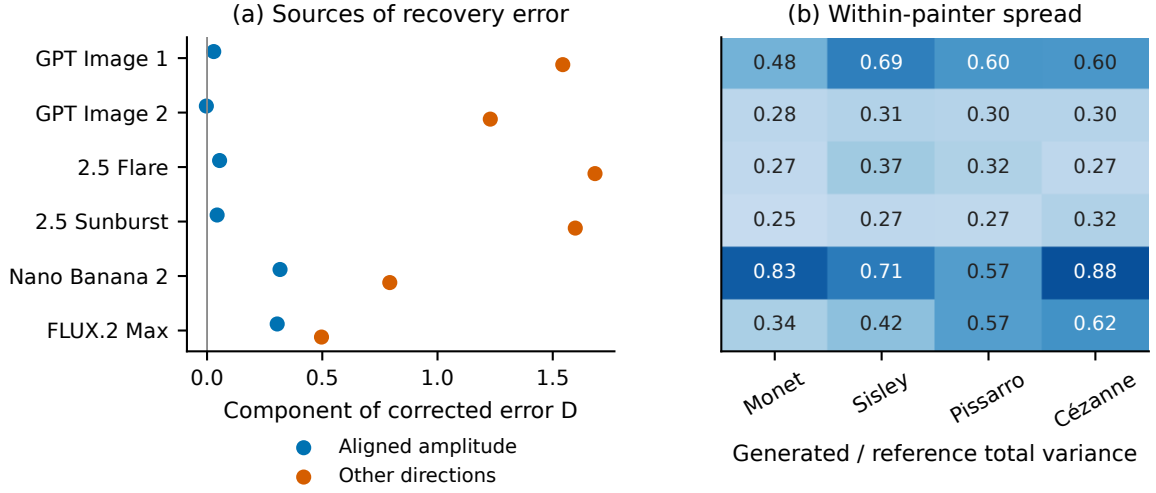


Figure 2: Complementary explanations of the measured mismatch. (a) Cross-repeat error separates reference-aligned amplitude from other artist-contrast directions. Small negative terms are retained. (b) Generated/reference covariance-trace ratios are below one for every model and painter. These total spreads include variation across the fixed briefs and do not identify within-scene mode collapse.

for Monet. These descriptive gains are neither uniform nor equivalent to artist-geometry recovery. Sunburst has the lowest observed named energy for Sisley and Cézanne, while FLUX has the lowest for Monet and Pissarro. In particular, Sunburst’s Cézanne energy falls from 2.399 under the generic clause to .745 under naming, despite its larger four-painter geometry error than FLUX. Reference-only projections illustrate the narrower, displaced clouds without establishing separation in all dimensions (Figure 3).

FLUX remains the smallest-error point estimate after each of 14 scene omissions. Common-square measurement, equal weighting of the four metadata-derived reference-content classes, and their combination retain the same two adjusted model differences. Class weighting changes the reference target, so absolute errors across targets are not directly comparable. Feature-family views also qualify the ranking: Nano Banana 2 has slightly lower texture-only error than FLUX. Removing texture brings Flare and Sunburst’s slopes close to one (.945 and .995), but their errors remain 1.580 and 1.604. The mismatch therefore persists outside texture coordinates. These sensitivity analyses neither validate the representation perceptually nor remove reference-content and capture differences.

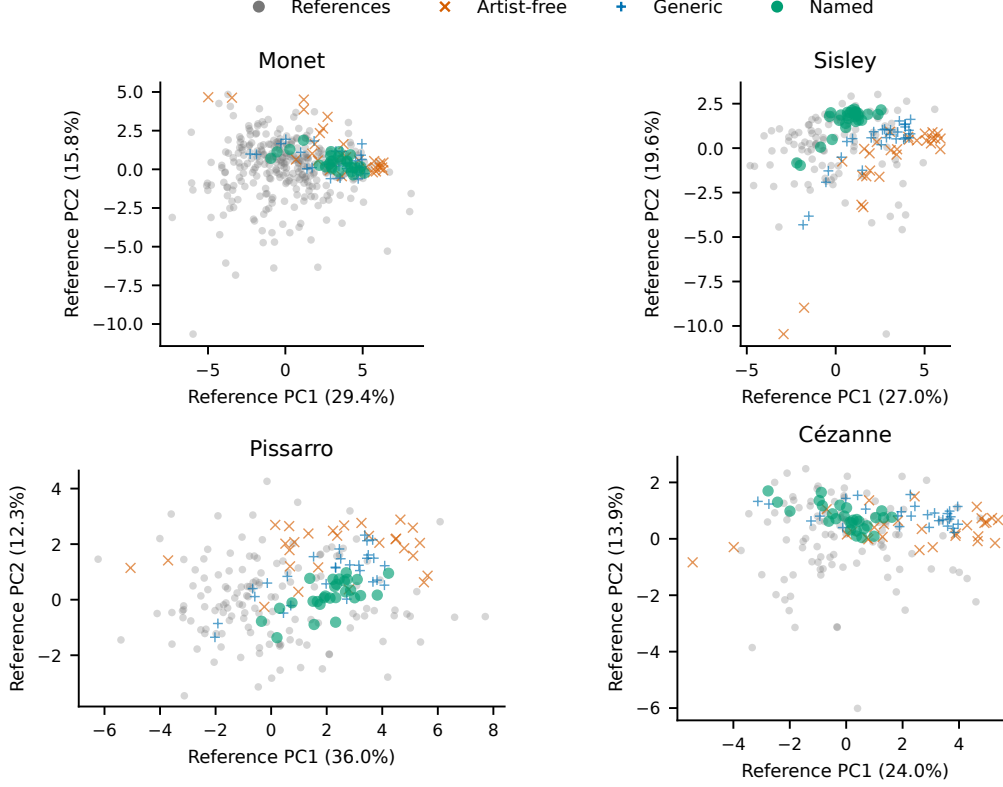


Figure 3: Reference and generated distributions for all four painters, illustrated with GPT Image 2.5 Sunburst. This model illustration was selected before new feature outcomes were inspected. Each panel uses principal components fitted only to that painter’s reference images, so axes differ between painters. All three generated clauses use the same 14 scenes and two repeats. These projections are descriptive; energy and spread use all 31 coordinates.

6 What the supporting interventions explain

6.1 Proximity gains do not establish painter specificity

An earlier exploration used the same four-painter reference panel with 1,536 named and 384 artist-free outputs from GPT Image 1 and GPT Image 2 request groups, three prompt methods and 16 scenes. Across its 24 painter–model–method cells, generated/reference trace ratios ranged from .206 to .376. Generated and reference images were also separable by a grouped classifier, with balanced accuracy .940–.983. These findings motivated the present experiment; they do not supply its inferential evidence or establish perceptual differences. The original projections remain available with the exploration’s numerical report.

The effect of naming on reference proximity was not uniform. In matched-size real/real comparisons, generated/reference energy exceeded the real/real median for every painter. Pooling the explored named conditions lowered the energy median relative to the free control only for Sisley. A later controlled Monet/Cézanne panel, with 24 shared scenes and three repeats, produced negative named-minus-free energy changes in all six model–painter comparisons; four rejected in their original eight-test family. The different reference panels, prompts and collection conditions preclude treating these as a single ranking.

A separate prospective Cézanne experiment supplied the stronger generic control: 96 outputs on 24 new scenes, with two repeats per clause. Naming reduced energy relative to the generic painting clause by 1.195 ($p = .00001$, fixed $\alpha = .025$), while the named/generic trace ratio was .552. Thus a generic clause did not reproduce the entire proximity gain in that comparison. This

is compatible with either correct or incorrect painter-specific contrasts; the current four-painter design directly distinguishes those cases.

6.2 Generic painting can alter responsiveness

Two separate palette experiments crossed six scenes with muted/vivid instructions and free, generic, Monet and Cézanne clauses, producing 192 outputs per collection. Their primary endpoint was the painter-minus-generic difference in the vivid-minus-muted response of median CIELAB chroma. Neither collection resolved an additional painter-name interaction: the original Monet and Cézanne estimates were $-.284$ and $-.018$ in development-IQR units, with simultaneous intervals $[-.585, .016]$ and $[-.343, .308]$; the later estimates were $-.159$ and $.037$, also with intervals spanning zero.

In contrast, the secondary generic-minus-free response was negative in both collections: $-.853$ and $-.891$, with nominal 95% intervals excluding zero. A generic painting clause can therefore attenuate the measured palette response without an identified additional name effect. This supports including a generic control rather than attributing every named/free difference to painter specificity. The unresolved interactions do not prove equivalence, and two palette levels cannot distinguish reduced response gain from a shifted operating range or saturation.

6.3 Moment-map explanations depend on the scene panel

Generated-only moment maps provide another proxy explanation. Translation matches the mean of named outputs; translation plus isotropic scaling also matches their total variance. In the earlier controlled panel, actual naming had lower held-scene reference energy than the translation/scaling map in all six model-painter averages. Mean and scalar spread changes therefore did not exhaust the observed naming effect. Adding scale improved prediction of named scene means in all six cells while worsening reference energy in five, showing that these two evaluation targets can disagree.

That historical ordering did not transfer to a prospective 240-image FLUX/Cézanne comparison on 12 new scenes. The unchanged translation/scaling map improved both reference energy and repeat-corrected scene-mean prediction relative to translation: differences $-.390$ and -5.047 , with approximate simultaneous intervals $[-.509, -.271]$ and $[-7.880, -2.214]$. The new scenes, collection time and finite-sample energy target differed. This contrary result rules out presenting the earlier ordering as a general mechanism. It motivates the present scene-conditional artist-contrast analysis, which evaluates recovery directly rather than identifying naming with a universal global transformation.

7 Discussion

The distinction between a common painting response and recovery of painter contrasts changes what counts as a successful model. Strong named/free movement is insufficient, as is recognition of the requested name. The relevant test is whether the differences generated by changing the name correspond to the differences in the reference artists. The reference-aligned slope and corrected error provide complementary answers: response amplitude alone cannot establish correct geometry, and mean geometry alone cannot establish full-distribution recovery.

The observed error is not explained by weak name conditioning alone. GPT Image 2 matches the amplitude of reference differences, yet its residual artist contrasts dominate the recovery error. The GPT Image 2.5 configurations likewise combine substantial aligned responses with larger off-axis errors and relatively little within-scene repeat dispersion. For these configurations, increasing the strength or consistency of the existing name response would not necessarily

repair the measured mismatch. FLUX.2 Max illustrates the opposite tradeoff: a weaker aligned response accompanies lower total error. This identifies the organization of artist contrasts, alongside response strength, as a target for model improvement.

The generic intervention supplies a partial explanation of the large common shift: its direction overlaps the shared named response, and the earlier palette experiments show that a generic painting clause can alter chroma responsiveness. It does not explain the entire shift or the residual painter contrasts. Nor does the decomposition identify separate latent modules in a generator. Training imbalance, artist–subject associations, preference optimization and inference-time guidance remain possible upstream causes. Distinguishing them would require interventions on training data, conditioning or model internals; the present results instead locate the mismatch in observable output geometry.

7.1 Scope of the evidence

Reference contrasts combine stylistic and content differences in this finite panel. Holding generated scene text fixed controls the prompt intervention but does not make historical content identical or verify that outputs follow every scene instruction. Results therefore concern recovery of measured reference painter differences under these outdoor briefs. They do not measure the complete style of an artist or establish generalization to unrelated artistic traditions. Four related painters make the common-response question useful but narrow its external scope.

The representation is transparent but not perceptually calibrated. A separate processing challenge found exact invariance for lossless round trips, while resizing and blur materially changed texture coordinates. Eight-neighbor local-binary-pattern entropy accounted for 83.9% of squared texture displacement under the tested blur. Common-square remeasurement retained all eight historical energy-contrast signs and the four adjusted rejections, but controls only one source of image geometry variation. Unknown photographic and reproduction processes remain potential contributors to generated/reference differences. No qualified independent-capture pairs or human style ratings are available.

Model comparisons apply to the requested configurations, clause template and collection period. Their repeat correction requires stable conditional responses and independent errors; randomized dispatch does not prove those conditions. The experiment has 14 scene clusters, so uncertainty may remain substantial despite many images. Reference-work resampling and scene deletions assess sensitivity without creating new independent replication. All collections were conducted by the same research team; there is no separate-investigator replication.

8 Conclusion

Painter-name prompting produces both a shared painting response and measurable reference-aligned artist variation. Across six generators, however, aligned response strength does not determine recovery of painter differences. Much of the observed error lies in other artist-contrast directions, even when the aligned amplitude is correct. FLUX.2 Max outperforms both GPT Image 2.5 variants on the specified geometry endpoint, without establishing a universal model ranking. Evaluating which differences a name induces, rather than only how much it moves an image toward a painter’s collection, reveals this limitation.

Data and code availability

Versioned protocols, exact prompts, model configurations, compact features, scalars and numerical replay code are retained with the analysis. The earlier experiments are available through the [project’s versioned releases](#). The new experiment’s compact inputs and exact replay commands

are retained in the accompanying project under `painter_specificity_v2`; a public release of this addition has not yet been archived. Numerical replay starts from retained feature vectors. Raw artwork, generated pixels and response bodies are excluded from the public packages; numerical replay therefore does not independently re-extract images or authenticate their acquisition. Source metadata records work identifiers, URLs and rights statements, without granting new rights to the images.

References

- Andrea Asperti. On the separation of human and AI-generated images in CLIP embedding space. arXiv:2608.25609, version 1, 2026. URL <https://arxiv.org/abs/2608.25609>. Preprint.
- Andrea Asperti, Franky George, Tiberio Marras, Razvan Ciprian Stricescu, and Fabio Zanotti. A critical assessment of modern generative models’ ability to replicate artistic styles. *Big Data and Cognitive Computing*, 9(9):231, 2025. doi: 10.3390/bdcc9090231.
- Yaniv Benny, Tomer Galanti, Sagie Benaim, and Lior Wolf. Evaluation metrics for conditional image generation. *International Journal of Computer Vision*, 129:1712–1731, 2021. doi: 10.1007/s11263-020-01424-w. URL <https://link.springer.com/article/10.1007/s11263-020-01424-w>.
- Adrien Delière, Jeanne Marlot, Marc Van Droogenbroeck, and Maria Giulia Dondero. How good is the machine at the imitation game? on stylistic characteristics of AI-generated images. *Journal of Imaging*, 11(12):429, 2025. doi: 10.3390/jimaging11120429. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC12734345/>.
- Jörn Diedrichsen and Nikolaus Kriegeskorte. Representational models: A common framework for understanding encoding, pattern-component, and representational-similarity analysis. *PLOS Computational Biology*, 13(4):e1005508, 2017. doi: 10.1371/journal.pcbi.1005508.
- Tarian Fu, Javier Conde, Gonzalo Martínez, Pedro Reviriego, Elena Merino-Gómez, and Fernando Moral. Artificial intelligence and misinformation in art: Can vision language models judge the hand or the machine behind the canvas? arXiv:2508.01408, version 1, 2025. URL <https://arxiv.org/abs/2508.01408v1>.
- D. Kim, S.-W. Son, and H. Jeong. Large-scale quantitative analysis of painting arts. *Scientific Reports*, 4:7370, 2014. doi: 10.1038/srep07370.
- J. Kim, B. Lee, T. You, and J. Yun. Context-aware multimodal AI navigates hidden pathways in five centuries of art evolution. *Proceedings of the National Academy of Sciences*, 123(30):e2517969123, 2026. doi: 10.1073/pnas.2517969123.
- B. Lee, M. K. Seo, D. Kim, I.-S. Shin, M. Schich, H. Jeong, and S. K. Han. Dissecting landscape art history with information theory. *Proceedings of the National Academy of Sciences*, 117(43):26580–26590, 2020. doi: 10.1073/pnas.2011927117.
- Stéphane G. Mallat. A theory for multiresolution signal decomposition: The wavelet representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11(7):674–693, 1989. doi: 10.1109/34.192463. URL <https://www.di.ens.fr/~mallat/papiers/MallatTheory89.pdf>.
- Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunje Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7176–7185, 2020. URL <https://proceedings.mlr.press/v119/naeem20a.html>.

- Timo Ojala, Matti Pietikäinen, and Topi Mäenpää. Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(7):971–987, 2002. doi: 10.1109/TPAMI.2002.1017623.
- Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL <https://arxiv.org/abs/2104.11222>.
- Gaurav Sharma, Wencheng Wu, and Edul N. Dalal. The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations. *Color Research and Application*, 30(1):21–30, 2005. doi: 10.1002/col.20070.
- H. Y. D. Sigaki, M. Perc, and H. V. Ribeiro. History of art paintings through the lens of entropy and complexity. *Proceedings of the National Academy of Sciences*, 115(37):E8585–E8594, 2018. doi: 10.1073/pnas.1800083115.
- Gowthami Somepalli, Anubhav Gupta, Kamal Gupta, Shramay Palta, Micah Goldblum, Jonas Geiping, Abhinav Shrivastava, and Tom Goldstein. Investigating style similarity in diffusion models. In *European Conference on Computer Vision*, 2024. URL https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/08294.pdf.
- Grace Su, Sheng-Yu Wang, Aaron Hertzmann, Eli Shechtman, Jun-Yan Zhu, and Richard Zhang. Identifying prompted artist names from generated images, 2025. URL <https://arxiv.org/abs/2507.18633v1>. Version 1.
- Gábor J. Székely and Maria L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013. doi: 10.1016/j.jspi.2013.03.018.
- Denny Vrandečić and Markus Krötzsch. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10):78–85, 2014. doi: 10.1145/2629489.
- Wikimedia Commons. Commons:GLAM, 2026. URL <https://commons.wikimedia.org/wiki/Commons:GLAM>. Accessed 11 September 2026.

A Feature definitions

Color coordinates use CIELAB under D65 and the 2° observer. Lightness and chroma each contribute a median and IQR. Chromatic fraction is the share of pixels with $C^* \geq 5$. Hue concentration is the magnitude of the chroma-weighted circular mean; hue entropy uses 24 equally spaced bins and is normalized by their log count. Both use chroma weights and are zero when fewer than 1% of pixels are chromatic. At each of three lags (1%, 4% and 16% of the short side), the median is taken over pooled horizontal and vertical CIEDE2000 differences. A log–log slope summarizes their scale dependence (Sharma et al., 2005).

Spatial and texture coordinates use linear-sRGB luminance. A Tukey window (parameter .1) precedes Fourier analysis. Thirty-two logarithmic radial bins cover 4–128 cycles per short side; a Theil–Sen line gives spectral slope and its root-mean-square residual. Power-weighted axial concentration measures anisotropy. Scharr derivatives supply gradient median and IQR, an 18-bin magnitude-weighted orientation entropy, and horizontal–vertical balance. Mean Jensen–Shannon divergence of quadrant orientation histograms from the global histogram measures regional variation.

Four log mean-squared detail energies come from a four-level undecimated db2 wavelet transform, followed by their linear slope and mean second difference (Mallat, 1989). Uniform local-binary-pattern entropy uses $(P, R) = (8, 1), (16, 2), (32, 4)$ on quantized luminance (Ojala et al., 2002). Three median local coefficients of variation use window widths obtained by rounding 1%, 4% and 16% of the short side and increasing even widths to the next odd integer, with a 10^{-6} mean floor. The fixed implementation records boundary handling and stabilizing constants. The separate scaler contains 101 Monet, 36 Sisley, 48 Pissarro and 36 Cézanne works. Equal painter mass is used to calculate each weighted median and IQR.

B Estimator details and sensitivity analyses

Let $\theta_{msa} = \mathbb{E}[d_{msak}]$ and $d_{msak} = \theta_{msa} + \eta_{msak}$. If repeat errors have zero mean and $\mathbb{E}\langle \eta_{msa1}, \eta_{msa2} \rangle = 0$, then

$$\mathbb{E}[\hat{D}_{ms}] = H^{-1} \sum_a \|\theta_{msa} - r_a\|^2.$$

Centering induces dependence between artists within a repeat, but independent repeat blocks preserve the vanishing cross-repeat term. A common response has $\theta_{msa} = 0$ and hence expected $D = 1$, even if it shifts the entire named collection toward the references. A correctly aligned but rescaled response $\theta_{msa} = br_a$ gives expected $D = (b - 1)^2$. Orthogonal deviations add their squared magnitude. This identity explains why more visibly distinct painter outputs need not have lower recovery error.

For each model, global distortion applies the same cross-product estimator to the scene-averaged contrasts. Its target differs from the mean scene-conditional error: averaging can cancel scene-dependent deviations. Both are reported, without selecting the smaller as the primary result. The primary comparison weights scenes and artists equally, irrespective of the reference count. The report also retains each painter’s additive contribution to D . These sum to the model error but are not standalone artist-fidelity scores, because the centering in Equation 1 couples the four named conditions.

For a model contrast, compute one error difference per complete paired scene. Its standard error is the sample standard deviation of those differences divided by $\sqrt{n}$. The simultaneous half-width is $t_{n-1, 1-.05/(2 \cdot 21)} \text{SE}$. Six slope intervals use the same family adjustment. Additional nominal intervals and 5,000 paired-scene bootstrap draws are descriptive checks, not alternative routes to significance. Reference sensitivity draws 1,000 bootstrap samples independently within each painter, recomputes reference means and H , and holds generated vectors and scaling fixed. It describes reference-panel dependence rather than all uncertainty jointly.

The four-class reference sensitivity uses a fixed lexicon applied to source titles: water-organized, built-place-organized, route-organized, and open or wooded land. Each class mean receives one-quarter weight within a painter. Resampling within artist/class strata conditions on those labels and does not account for classification error. The older controlled naming panel used a separate three-class visual coding scheme, described below.

The shared-effect diagnostic first averages each arm over scenes within repeat. Let h_{ak} be the named-minus-free difference and $c_k = \frac{1}{4} \sum_a h_{ak}$. The cross-repeat common and artist-specific squared magnitudes are $4\langle c_1, c_2 \rangle$ and $\sum_a \langle h_{a1} - c_1, h_{a2} - c_2 \rangle$. The common fraction divides the first magnitude by their sum and is reported only when that sum is positive; finite corrected components can be negative, so the ratio is not a probability. Generic/common cosine uses their repeat-averaged vectors. This descriptive global statistic is separate from the conditional recovery endpoint.

C Complete model and painter summaries

Table 3 reports the full model-comparison family; Table 4 retains distributional outcomes for every painter and model. The latter do not inherit the primary model-contrast significance tests. Their energy values concern empirical distributions of 28 named or generic outputs against the recorded reference panel.

Model A	Model B	$D_A - D_B$ [simultaneous interval]
GPT Image 1	GPT Image 2	0.346 [−0.400, 1.093]
GPT Image 1	2.5 Flare	−0.165 [−1.027, 0.697]
GPT Image 1	2.5 Sunburst	−0.069 [−1.181, 1.043]
GPT Image 1	Nano Banana 2	0.463 [−0.602, 1.528]
GPT Image 1	FLUX.2 Max	0.771 [−0.037, 1.580]
GPT Image 2	2.5 Flare	−0.512 [−1.150, 0.126]
GPT Image 2	2.5 Sunburst	−0.415 [−1.181, 0.350]
GPT Image 2	Nano Banana 2	0.116 [−0.922, 1.155]
GPT Image 2	FLUX.2 Max	0.425 [−0.161, 1.011]
2.5 Flare	2.5 Sunburst	0.096 [−0.286, 0.478]
2.5 Flare	Nano Banana 2	0.628 [−0.283, 1.540]
2.5 Flare	FLUX.2 Max	0.937 [0.256, 1.618]
2.5 Sunburst	Nano Banana 2	0.532 [−0.436, 1.500]
2.5 Sunburst	FLUX.2 Max	0.840 [0.058, 1.623]
Nano Banana 2	FLUX.2 Max	0.309 [−0.847, 1.464]

Table 3: All 15 paired model comparisons in the prespecified inferential family. A negative contrast favors model A on the stated feature-geometry target. Flare and Sunburst denote the corresponding GPT Image 2.5 models.

Model	Monet	Sisley	Pissarro	Cézanne
GPT Image 1	−2.515 / 0.479	−2.739 / 0.694	−2.628 / 0.603	−1.336 / 0.598
GPT Image 2	−0.602 / 0.283	−1.092 / 0.312	−0.195 / 0.303	−0.373 / 0.303
2.5 Flare	0.962 / 0.269	−0.008 / 0.371	0.429 / 0.321	−1.013 / 0.275
2.5 Sunburst	0.267 / 0.248	−0.718 / 0.274	−0.091 / 0.273	−1.653 / 0.318
Nano Banana 2	−0.513 / 0.829	−1.381 / 0.708	−0.741 / 0.572	−1.102 / 0.877
FLUX.2 Max	−0.707 / 0.340	−0.886 / 0.416	−1.080 / 0.570	−1.431 / 0.622

Table 4: Complementary outcomes for all four painters: each cell gives named-minus-generic reference energy / generated-to-reference total variance. These are descriptive empirical distribution summaries, without additional tests. Lower energy change favors naming over the generic clause; a variance ratio below one indicates a narrower generated feature distribution.

D Supporting cohorts and interpretation

The four-painter exploration used repeated scenes and three prompt methods. Its projected distributions, trace ratios and grouped discrimination are post-result descriptions. The pooled named energy medians for Monet, Sisley, Pissarro and Cézanne were 2.068, 1.678, 1.975 and 2.142, compared with free medians 1.732, 1.878, 1.854 and 1.657. Matched-size real/real medians were .193, .215, .164 and .199. These summaries neither isolate a prompt method nor identify a generator mechanism.

The later controlled naming panel selected Monet and Cézanne after the exploration. It crossed 24 scenes with three models, two clauses and three repeats; 142 additional short-scene outputs supplied two prompt-length comparisons. Reference panels contain 38 Monet and 32

Cohort	Comparison	Role in this paper
Four-painter exploration	649 references; 1,920 generated outputs	Describes reference/generated separation and motivates the controlled test.
Controlled naming	1,006 outputs; 38 Monet and 32 Cézanne references	Six named/free and two detailed/short prompt contrasts; subsequent transformation and conditional-variance diagnostics.
Palette interventions	Two separate 192-output collections	Named-minus-generic chroma response; no pooling across collections.
Generic-clause comparison	96 Cézanne/generic outputs on 24 new scenes	Tests a proximity gain beyond a generic painting clause.
Fixed-map transfer	240 FLUX/Cézanne outputs on 12 new scenes	Prospective counterexample to the historical opposing energy/prediction ordering.

Table 5: Supporting evidence is kept separate from the new six-model experiment. Different prompts, panels and finite-sample targets do not form one pooled test.

Cézanne works. Their water/built/land labels came from maintainer-run LLM visual coding. The generated observations were weighted to these reference class proportions in the controlled comparisons. The new primary artist-contrast analysis does not use class weights.

The six controlled named-minus-free energy changes were $-.846, -1.818$ for Nano Banana 2, $-.625, -.896$ for FLUX.2 Max, and $-.234, -.056$ for GPT Image 2 (Monet, Cézanne). The first four rejected after adjustment in the original family of eight; the two GPT Image 2 contrasts did not. A separate 72-output FLUX follow-up repeated the directions with changes $-.695, -.952$ and adjusted rejection in its new four-endpoint family. These are repeated collections by the same investigators, not independent-investigator replication.

The standalone generic-clause comparison randomized its two clauses within 48 two-position blocks, two per scene. It used the 32-work Cézanne panel and water/built/land masses (3, 11, 18)/32 for each generated clause. Its two-sided Monte Carlo test used 99,999 sign draws with the plus-one correction and a fixed threshold of .025. The test assumes exchangeability under within-block clause swaps: it tests a sharp no-effect null, not equality of energy discrepancies alone.

Conditional variance and scene retrieval need not move together. In the controlled FLUX/Monet panel, naming reduced between-scene spread but increased held-repetition scene retrieval from 44.4% to 59.7%. Retrieval assigns each held-out image to the nearest centroid formed from the other repetitions; it measures relative feature distinguishability, not verified scene adherence. This example prevents equating contraction with loss of all content response.

The palette experiments used six fixed scenes and four repeats per arm/polarity. Original approximate intervals used a two-interaction family; the later family also included two FLUX naming endpoints. Their unresolved name-minus-generic estimates do not establish equality between names and a generic clause. Secondary generic-minus-free intervals were nominal.

Historical transformation maps were fitted to generated outputs without reference optimization and evaluated on held-out whole scenes. The prospective fixed-map experiment used the original full fits unchanged, 12 new scenes and ten repetitions. Its energy target depends on that repeat count; conditional prediction error uses a repeat-noise correction. The two simultaneous jackknife intervals assume stable independent repeat-block errors. The observed conditional-error half-width exceeded its finite-proxy planning criterion, so planning calculations did not guarantee achieved precision. The contrary prospective result is retained without extra sampling.

E Collection provenance and computational scope

Historical GPT Image 1 and GPT Image 2 labels identify the requested model names. A later technical negative control showed that the earlier image interface also accepted an invented image-model identifier. This does not show that valid names select the same underlying model, but successful request acceptance alone cannot authenticate their separation. Historical contrasts are therefore interpreted as recorded request-group comparisons. The current experiment uses explicit documented model identifiers and fixed routes; it does not reuse historical images as verified examples of the new variants. All current outputs decode to the requested square dimensions, but their response bodies do not echo model or quality fields. Labels therefore denote the documented request configurations, without independent checkpoint attestation. Underlying checkpoints and training corpora are not available for inspection.

The present experiment excludes technical probes and an earlier attempt closed before feature measurement because model selection could not be qualified. No image is shared between these collections. The earlier incomplete generic-clause cohort is likewise not pooled with its complete 96-image successor. Exact assignments, attempt outcomes and measurement exclusions are retained in the versioned manifests; selection was not based on visual success.